22521-2022-Winter-model-answer-paper[Msbte study resources]

computer engineering syco (Vidya Vikas Pratishthan Institute Of Engineering & Technology)

WINTER – 2022 EXAMINATION
Model Answer

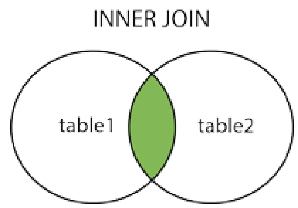
Subject Name: **Advanced Database Management System** **Subject Code:** 22521

Important Instructions to examiners:

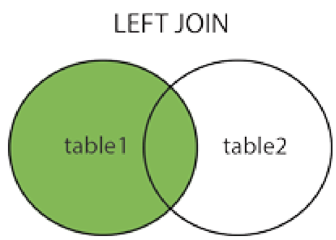
1) The answers should be examined by key words and not as word-to-word as given in the model answer scheme.
2) The model answer and the answer written by candidate may vary but the examiner may try to assess the understanding level of the candidate.
3) The language errors such as grammatical, spelling errors should not be given more Importance (Not applicable for subject English and Communication Skills.
4) While assessing figures, examiner may give credit for principal components indicated in the figure. The figures drawn by candidate and model answer may vary. The examiner may give credit for any equivalent figure drawn.
5) Credits may be given step wise for numerical problems. In some cases, the assumed constant values may vary and there may be some difference in the candidate's answers and model answer.
6) In case of some questions credit may be given by judgement on part of examiner of relevant answer based on candidate's understanding.
7) For programming language papers, credit may be given to any other program based on equivalent concept.
8) As per the policy decision of Maharashtra State Government, teaching in English/Marathi and Bilingual (English + Marathi) medium is introduced at first year of AICTE diploma Programme from academic year 2021-2022. Hence if the students in first year (first and second semesters) write answers in Marathi or bilingual language (English +Marathi), the Examiner shall consider the same and assess the answer based on matching of concepts with model answer.

Q. No.	Sub Q. N.	Answer	Marking Scheme
1		**Attempt any FIVE of the following:**	**10 M**
	a)	**State the use of concurrency control.**	**2 M**
	Ans	It is a procedure in DBMS which helps us for the management of two simultaneous processes to execute without conflicts between each other, these conflicts occur in multi user systems. Concurrency can simply be said to be executing multiple transactions at a time. It is required to increase time efficiency. If many transactions try to access the same data, then inconsistency arises. Concurrency control required to maintain consistency data. • Waiting time will be decreased. • Response time will decrease. • Resource utilization will increase. • System performance & Efficiency is increased.	Any 2 uses- 2M

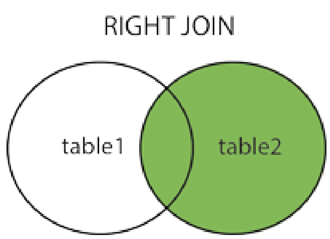
	b)	**Define Big Data.**	**2 M**
	Ans	Big Data contains a large amount of data that is not being processed by traditional data storage or the processing unit. It is used by many **multinational companies** to **process** the data and business of many **organizations**. The data flow would exceed **150 exabytes** per day before replication. There are five v's of Big Data that explains the characteristics. 5 V's of Big Data Volume (huge amount of data), Veracity (inconsistencies and uncertainty of data), Variety (different formats of data from various sources), Value (extract useful data), Velocity (high speed of accumulation of data)	Correct definition-2M
	c)	**Give any four characteristics of XML.**	**2 M**
	Ans	XML separates data from HTML XML simplifies data sharing XML simplifies data transport XML simplifies Platform change XML increases data availability XML can be used to create new internet languages	**four characteristics of XML-2M**
	d)	**Define Data Mart & Meta Data.**	**2 M**
	Ans	**Data Mart**: A data mart is **a subset of a data warehouse focused on a particular line of business, department, or subject area**. Data marts make specific data available to a defined group of users, which allows those users to quickly access critical insights without wasting time searching through an entire data warehouse. **Meta Data**: Metadata in DBMS is characterized as **data about data**. It describes the context and information about data, the way that data is stored and the various relations among data. Metadata in a relational DBMS stores data about Constraints, Table Relationships, Data Types, Columns, Tables, and so on.	**Data Mart-1M** **& Meta Data-1M**
	e)	**Enlist any four types of join.**	**2 M**
	Ans	• Inner Join : Returns records that have matching values in both tables	Listing of 4 join types-2M(1/2 M each)



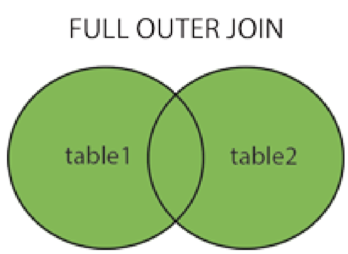
- Left Outer Join: Returns all records from the left table, and the matched records from the right table



- Right Outer Join: Returns all records from the right table, and the matched records from he left table



- Full Join / Full Outer Join: Returns all records when there is a match in either left or right table



	f)	**Enlist Data Mining Techniques.**	**2 M**
	Ans	**Following are the Data Mining Techniques:** Classification, Clustering, Regression, Prediction, Association rules, Sequential patterns, Data cleaning, Data visualization, machine learning.	Any 4 -2M (1/2 M each)
	g)	**Write any four benefits of NOSQL.**	**2 M**

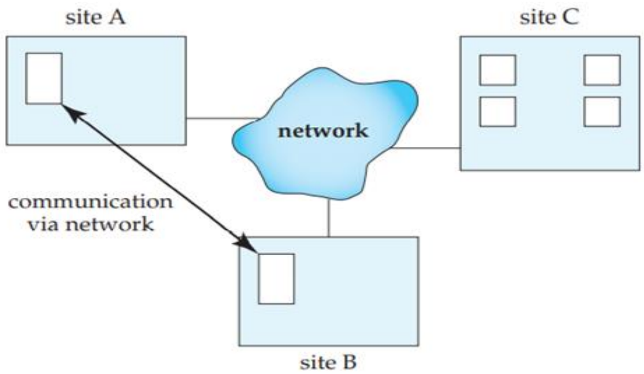
	Ans	**Flexible Data Model:** NoSQL databases are highly flexible as they can store and combine any type of data, both structured and unstructured, unlike relational databases that can store data in a structured way only. **Evolving Data Model:** NoSQL databases allow you to dynamically update the schema to evolve with changing requirements while ensuring that it would cause no interruption or downtime to your application **Elastic Scalability:** NoSQL databases can scale to accommodate any type of data growth while maintaining low cost. **High Performance:** NoSQL databases are built for great performance, measured in terms of both throughput (it is a measure of overall performance) and latency (it is the delay between request and actual response). **Open-source:** NoSQL databases don't require expensive licensing fees and can run on inexpensive hardware, rendering their deployment cost-effective.	Any 4 benefits-2M (1/2 M each)
2.		**Attempt any <u>THREE</u> of the following:**	**12 M**
	a)	**Explain nested query with suitable example.**	**4 M**
	Ans	• A nested query is a query that has another query embedded within it. • The embedded query is called a subquery. • A subquery typically appears within the WHERE clause of a query. • It can sometimes appear in the FROM clause or HAVING clause. • A nested query in SQL contains a query inside another query. • The result of the inner query will be used by the outer query. • Example: Select all Developers who earn more than all the Managers. SELECT * FROM employees WHERE role = 'Developer' AND salary > ALL (SELECT salary FROM employees WHERE role = 'Manager');	Explnation-3M Example-1M
	b)	**Differentiate between SQL & NOSQL database system.**	**4 M**

	Ans		Any 4 ponts-4M

SQL	NoSQL
Databases are categorized as Relational Database Management System (RDBMS).	NoSQL databases are categorized as Non-relational or distributed database system.
SQL databases have fixed or static or predefined schema.	NoSQL databases have dynamic schema.
SQL databases display data in form of tables so it is known as table-based database.	NoSQL databases display data as collection of key-value pair, documents, graph databases or wide-column stores.
SQL databases are vertically scalable.	NoSQL databases are horizontally scalable.
SQL databases use a powerful language "Structured Query Language" to define and manipulate the data.	In NoSQL databases, collection of documents are used to query the data. It is also called unstructured query language. It varies from database to database.
SQL databases are best suited for complex queries.	NoSQL databases are not so good for complex queries because these are not as powerful as SQL queries.
SQL databases are not best suited for hierarchical data storage.	NoSQL databases are best suited for hierarchical data storage.
MySQL, Oracle, Sqlite, PostgreSQL and MS-SQL etc. are the example of SQL database.	MongoDB, BigTable, Redis, RavenDB, Cassandra, Hbase, Neo4j, CouchDB etc. are the example of nosql database.

	c)	**Explain distributed database in detail.**	**4 M**
	Ans	• **Distributed database** is a database that is not limited to one computer system. It is like a database that consists of two or more files located in different computers or sites either on the same network or on an entirely different network. • These sites do not share any physical component. Distributed databases are needed when a particular data in the database needs to be accessed by various users globally. It needs to be handled in such a way that for a user it always looks like one single database. • By contrast, a **Centralized database** consists of a single database file located at one site using a single network.	Proper explanation with diagram-4M

- Logically connected single database and that will be dispersed to different sites/ nodes (connected by computer communication network).



Features of Distributed Databases

In general, distributed databases include the following features:

- **Location independency**: Data is independently stored at multiple sites and managed by independent Distributed database management systems **(DDBMS)**.
- **Network linking**: All distributed databases in a collection are linked by a network and communicate with each other.
- **Distributed query processing**: Distributed query processing is the procedure of answering queries (which means mainly read operations on large data sets) in a distributed environment.
- **Hardware independent**: The different sites where data is stored are hardware-independent. There is no physical contact between these distributed databases which is accomplished often through virtualization.

Distributed transaction management: Distributed database provides a consistent distribution through commit protocols, distributed recovery methods, and distributed concurrency control techniques in case of many transaction failures.

Distributed Data Storage

Consider a relation r that is to be stored in the database. There are two approaches to storing this relation in the distributed database:

- **Replication**. The system maintains several identical replicas (copies) of the relation, and stores each replica at a different site. The alternative to replication is to store only one copy of relation r.
- **Fragmentation**. The system partitions the relation into several fragments, and stores each fragment at a different site.

	d)	**Explain Aggregation in MapReduce.**	**4 M**
	Ans	• Map-reduce is a data processing programming model that helps to perform operations on large data sets and produce aggregated results. • MongoDB provides the mapReduce() function to perform the map-reduce operations. This function has two main functions, i.e., map function and reduce function. • The map function is used to group all the data based on the key-value and the reduce function is used to perform operations on the mapped data. • So, the data is independently mapped and reduced in different spaces and then combined together in the function and the result will save to the specified new collection. • This mapReduce() function generally operated on large data sets only. • Using Map Reduce you can perform aggregation operations such as max, avg on the data using some key and it is similar to groupBy in SQL. • It performs on data independently and parallel. In this example, we have five records from which we need to take out the maximum marks of each section and the keys are id, sec, marks. {"id":1, "sec":A, "marks":80} {"id":2, "sec":A, "marks":90} {"id":1, "sec":B, "marks":99} {"id":1, "sec":B, "marks":95} {"id":1, "sec":C, "marks":90} Here we need to find the maximum marks in each section. So, our key by which we will group documents is the sec key and the value will be marks. Inside the map function, we use emit(this.sec, this.marks) function, and we will return the sec and marks of each record(document) from the emit function. This is similar to group By MySQL. var map = function(){emit(this.sec, this.marks)}; After iterating over each document Emit function will give back the data like this: {“A”:[80, 90]}, {“B”:[99, 90]}, {“C”:[90] } var reduce = function(sec,marks){return Array.max(marks);};	Proper explanation with example- 4M

		Here in reduce() function, we have reduced the records now we will output them into a new collection. {out :"collectionName"} db.collectionName.mapReduce(map,reduce,{out :"collectionName"}); In the above query we have already defined the map, reduce.	
3.		**Attempt any <u>THREE</u> of the following:**	**12 M**
	a)	**With neat diagram explain Data warehousing Lifecycle.**	**4 M**
	Ans	 1. **Requirement Specification:** It is the first step in the development of the Data Warehouse and is done by business analysts. In this step, Business Analysts prepare business requirement specification documents. More than 50% of requirements are collected from the client side and it takes 3-4 months to collect all the requirements. After the requirements are gathered, the data modeler starts recognizing the dimensions, facts & combinations based on the requirements. We can say that this is an overall blueprint of the data warehouse. But, this phase is more about determining business needs and placing them in the data warehouse. 2. **Data Modelling:** This is the second step in the development of the Data Warehouse. Data Modelling is the process of visualizing data distribution and designing databases by fulfilling the requirements to transform the data into a format that can be stored in the data warehouse. For example, whenever we start building a house, we put all the things in the correct position as specified in the blueprint.	1 M for diagram and 3 M for explanation

		3. **ELT Design and Development:** This is the third step in the development of the Data Warehouse. ETL or Extract, Transfer, Load tool may extract data from various source systems and store it in a data lake. An ETL process can extract the data from the lake, after that transform it and load it into a data warehouse for reporting. For optimal speeds, good visualization, and the ability to build easy, replicable, and consistent data pipelines between all of the existing architecture and the new data warehouse, we need ELT tools. 4. **OLAP Cubes:** This is the fourth step in the development of the Data Warehouse. An OLAP cube, also known as a multidimensional cube or hypercube, is a data structure that allows fast analysis of data according to the multiple dimensions that define a business problem. 5. **UI Development:** This is the fifth step in the development of the Data Warehouse. So far, the processes discussed have taken place at the backend. There is a need for a user interface for how the user and a computer system interact, in particular the use of input devices and software, to immediately access the data warehouse for analysis and generating reports. 6. **Maintenance:** This is the sixth step in the development of the Data Warehouse. In this phase, we can update or make changes to the schema and data warehouse's application domain or requirements. Data warehouse maintenance systems must provide means to keep track of schema modifications as well, for instance, modifications. At the schema level, we can perform operations for the Insertion, and change dimensions and categories. Changes are, for example, adding or deleting user-defined attributes. 7. **Test and Deployment:** This is often the ultimate step in the Data Warehouse development cycle. Businesses and organizations test data warehouses to ensure whether the required business problems are implemented successfully or not. The warehouse testing involves the scrutiny of enormous volumes of data.	
	b)	**Explain Multimedia Databases.**	**4 M**
	Ans	Multimedia database is a collection of multimedia data which includes text, images, graphics (drawings, sketches), animations, audio, and video, among others. These databases have extensive amounts of data which can be multimedia and multisource. The framework which manages these multimedia databases and their different types so that the data can be stored, utilized, and delivered in more than one way is known as a multimedia database management system. The multimedia database can be classified into three types. These types are: 1. Static media 2. Dynamic media 3. Dimensional media	4 M for explanation

The contents of a multimedia database management system can be:

1. **Media data:** It is the actual data which represents an object.
2. **Media format data:** The information such as resolution, sampling rate, encoding system, etc. about the format of the media data under consideration after is undergoes acquisition, processing, and encoding is the media format data.
3. **Media keyword data:** Media keyword data are the keyword description related to the generation of data. This data is also known as content descriptive data. Examples of content descriptive data are place, time, date of recording.
4. **Media feature data:** Media feature data contains data which is content dependent such as kind of texture, distribution of, and the different shapes present in the data.

c) **Explain with example any four operation with MongoDB.** **4 M**

Ans

1 M for each correct operation

(4 operation should explained)

1. **Create Operations** –The create or insert operations are used to insert or add new documents in the collection. If a collection does not exist, then it will create a new collection in the database. You can perform, create operations using the following methods provided by the MongoDB:

Method	**Description**
db.collection.insertOne()	It is used to insert a single document in the collection.
db.collection.insertMany()	It is used to insert multiple documents in the collection.
db.createCollection()	It is used to create an empty collection.

Example: In this example, we are inserting details of a single student in the form of document in the student collection using db.collection.insertOne () method.

```
anki — mongo — 80×55
> use GeeksforGeeks
switched to db GeeksforGeeks
> db.student.insertOne({
... name : "Sumit",
... age : 20,
... branch : "CSE",
... course : "C++ STL",
... mode : "online",
... paid : true,
... amount : 1499
... })
{
        "acknowledged" : true,
        "insertedId" : ObjectId("5e540cdc92e6dfa3fc48ddae")
}
>
```

2. **Read Operations** –The Read operations are used to retrieve documents from the collection, or in other words, read operations are used to query a collection for a document. You can perform read operation using the following method provided by the MongoDB:

Method	Description
db.collection.find()	It is used to retrieve documents from the collection.

Example: In this example, we are retrieving the details of students from the student collection using db.collection.find() method.

```
anki — mongo — 80×55
> use GeeksforGeeks
switched to db GeeksforGeeks
> db.student.find().pretty()
{
        "_id" : ObjectId("5e540cdc92e6dfa3fc48ddae"),
        "name" : "Sumit",
        "age" : 20,
        "branch" : "CSE",
        "course" : "C++ STL",
        "mode" : "online",
        "paid" : true,
        "amount" : 1499
}
{
        "_id" : ObjectId("5e540d3192e6dfa3fc48ddaf"),
        "name" : "Sumit",
        "age" : 20,
        "branch" : "CSE",
        "course" : "C++ STL",
        "mode" : "online",
        "paid" : true,
        "amount" : 1499
}
{
        "_id" : ObjectId("5e540d3192e6dfa3fc48ddb0"),
        "name" : "Rohit",
        "age" : 21,
        "branch" : "CSE",
        "course" : "C++ STL",
        "mode" : "online",
        "paid" : true,
        "amount" : 1499
}
>
```

3. **Update Operations** –The update operations are used to update or modify the existing document in the collection. You can perform update operations using the following methods provided by the MongoDB

Method	Description
db.collection.updateOne()	It is used to update a single document in the collection that satisfy the given criteria.
db.collection.updateMany()	It is used to update multiple documents in the collection that satisfy the given criteria.
db.collection.replaceOne()	It is used to replace single document in the collection that satisfy the given criteria.

Example: In this example, we are updating the year of course in all the documents in the student collection using db.collection.updateMany() method.

```
anki — mongo — 80×43
> use GeeksforGeeks
switched to db GeeksforGeeks
> db.student.updateMany({}, {$set: {year: 2020}})
{ "acknowledged" : true, "matchedCount" : 3, "modifiedCount" : 3 }
> db.student.find().pretty()
{
        "_id" : ObjectId("5e540cdc92e6dfa3fc48ddae"),
        "name" : "Sumit",
        "age" : 24,
        "branch" : "CSE",
        "course" : "C++ STL",
        "mode" : "online",
        "paid" : true,
        "amount" : 1499,
        "year" : 2020
}
{
        "_id" : ObjectId("5e540d3192e6dfa3fc48ddaf"),
        "name" : "Sumit",
        "age" : 20,
        "branch" : "CSE",
        "course" : "C++ STL",
        "mode" : "online",
        "paid" : true,
        "amount" : 1499,
        "year" : 2020
}
{
        "_id" : ObjectId("5e540d3192e6dfa3fc48ddb0"),
        "name" : "Rohit",
        "age" : 21,
        "branch" : "CSE",
        "course" : "C++ STL",
        "mode" : "online",
        "paid" : true,
        "amount" : 1499,
        "year" : 2020
}
>
```

4. **Delete Operations** –The delete operation are used to delete or remove the documents from a collection. You can perform delete operations using the following methods provided by the MongoDB:

Method	Description
db.collection.deleteOne()	It is used to delete a single document from the collection that satisfy the given criteria.
db.collection.deleteMany()	It is used to delete multiple documents from the collection that satisfy the given criteria.

Example: In this example, we are deleting a document from the student collection using db.collection.deleteMany() method.

anki — mongo — 80×56

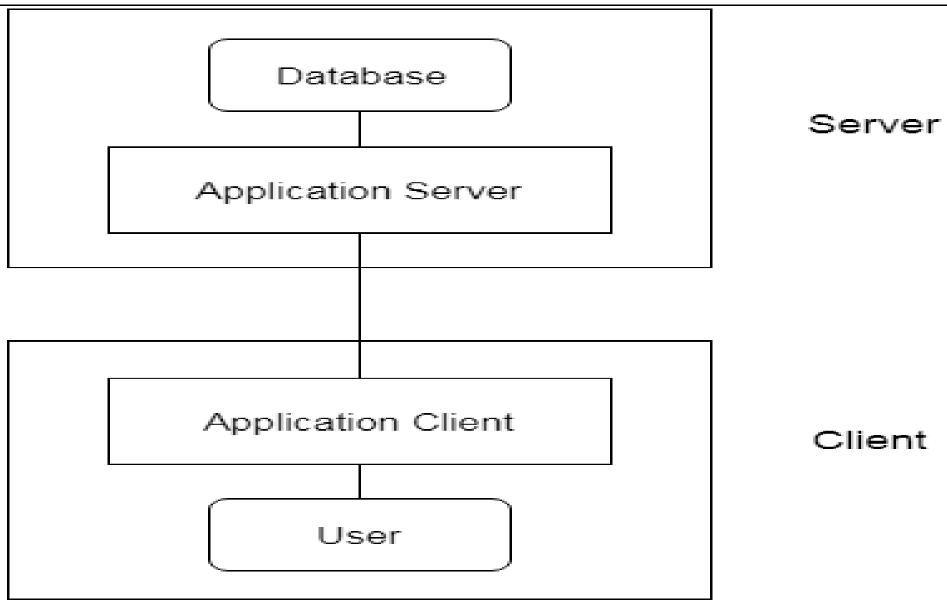
```
> use GeeksforGeeks
switched to db GeeksforGeeks
> db.student.find().pretty()
{
        "_id" : ObjectId("5e540d3192e6dfa3fc48ddaf"),
        "name" : "Sumit",
        "age" : 20,
        "branch" : "CSE",
        "course" : "C++ STL",
        "mode" : "online",
        "paid" : true,
        "amount" : 1499,
        "year" : 2020
}
{
        "_id" : ObjectId("5e54103592e6dfa3fc48ddb1"),
        "name" : "Rohit",
        "age" : 21,
        "branch" : "CSE",
        "course" : "C++ STL",
        "mode" : "online",
        "paid" : true,
        "amount" : 1499
}
> db.student.deleteMany({})
{ "acknowledged" : true, "deletedCount" : 2 }
>
```

	d)	**Explain features of R-programming along with its R-programming use in various applications.**	**4 M**

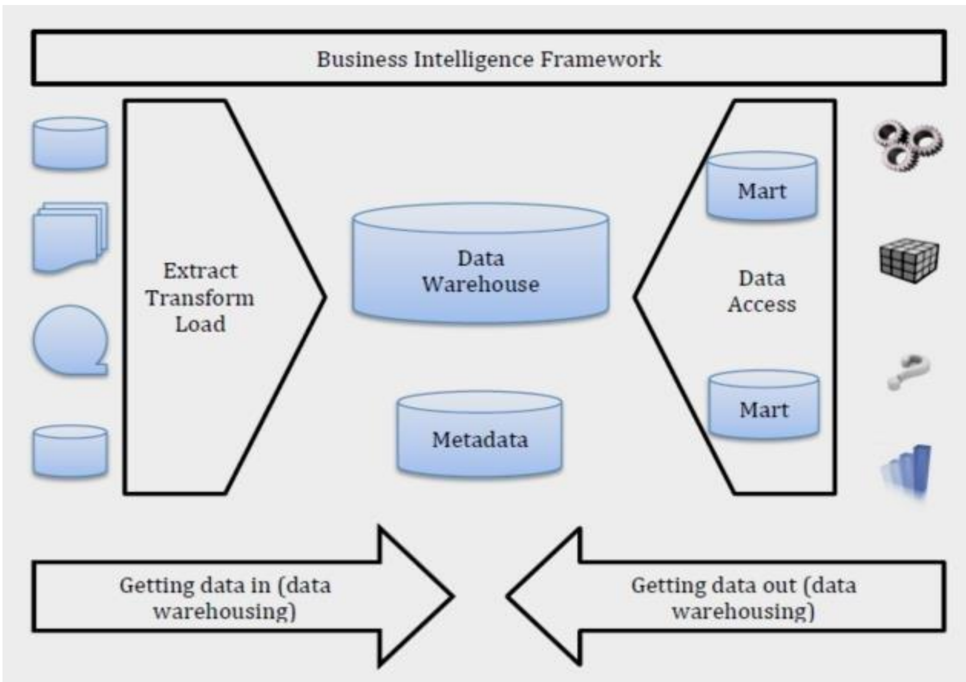
	Ans	**1.** Comprehensive Language-R is a comprehensive programming language, meaning that it provides services for statistical modeling as well as for software development. **2.** Provides a Wide Array of Packages-R is most widely used because of its wide availability of libraries. R has CRAN, which is a repository holding more than 10, 0000 packages. These packages appeal to every functionality and different fields that deal with data. Based on user requirements and preferences, these packages provide different features to their users. **3.** Possesses a Number of Graphical Libraries-The most important feature of R that sets it apart from other programming languages of Data Science is its massive collection of graphical libraries like ggplot2, plotly, etc. that are capable of making aesthetic and quality visualizations. **4.** Open-source-R is an open-source programming language. This means that it is free of cost and requires no license. Furthermore, you can contribute towards the development of R, customize its packages and add more features. **5.** Cross-Platform Compatibility-R supports cross-platform compatibility. It can be run on any OS in any software environment. It can also be run on any hardware configuration without any extra workarounds. **6.** Facilities for Various Industries-Almost every industry that makes use of data, utilizes the R language. While only the academic areas made use of R in the past, it is now being heavily used in industries that require to mine insights from the data. **7.** Performs Fast Calculations-Through R, you can perform a wide variety of complex operations on vectors, arrays, data frames and other data objects of varying sizes. Furthermore, all these operations operate at a lightning speed. It provides various suites of operators to perform these miscellaneous calculations. **8.** Integration with Other Technologies-R can be integrated with a number of different technologies, frameworks, software packages, and programming languages. It can be paired with Hadoop to use its distributed computing ability. It can also be integrated with programs in other programming languages like C, C++, Java, Python, and FORTRAN.	1 M for each feature (4 features are expected)
4.		**Attempt any <u>THREE</u> of the following:**	**12 M**
	a)	**Explain Mobile database with its need.**	**4 M**
	Ans	A Mobile database is a database that can be connected to a mobile computing device over a mobile network (or wireless network). Here the client and the server have wireless connections. In today's world, mobile computing is growing very rapidly, and it is huge potential in the field of the database. It will be applicable on different-different devices	2 M for explanation and 2 M for

<table>
<tr><td></td><td></td><td>like android based mobile databases, iOS based mobile databases, etc. Common examples of databases are Couch base Lite, Object Box, etc.

Need of Mobile database:
A cache is maintained to hold frequent and transactions so that they are not lost due to connection failure.
As the use of laptops, mobile and PDAs is increasing to reside in the mobile system.
Mobile databases are physically separate from the central database server.
Mobile databases resided on mobile devices.
Mobile databases are capable of communicating with a central database server or other mobile clients from remote sites.
With the help of a mobile database, mobile users must be able to work without a wireless connection due to poor or even non-existent connections (disconnected).
A mobile database is used to analyze and manipulate data on mobile devices.</td><td>need of mobile database</td></tr>
<tr><td></td><td>b)</td><td>Explain Three-Tier client server model.</td><td>4 M</td></tr>
<tr><td></td><td>Ans</td><td>



The 3-Tier architecture contains another layer between the client and server. In this architecture, client can't directly communicate with the server.
The application on the client-end interacts with an application server which further communicates with the database system.
End user has no idea about the existence of the database beyond the application server. The database also has no idea about any other user beyond the application.
The 3-Tier architecture is used in case of large web application.</td><td>2 M for diagram and 2 M for explanation</td></tr>
<tr><td></td><td>c)</td><td>Describe Business Intelligence Framework & Features.</td><td>4 M</td></tr>
<tr><td></td><td>Ans</td><td>Business intelligence (BI) refers to a technology-driven process that transforms the data into actionable information. Organizations have a huge flow of data coming from their customer end.
The data-driven decisions help in decision-making purposes in the organization.</td><td>2 M for BI framework</td></tr>
</table>

		• It has a large dataset processed on relational databases. • It represents results on dashboards and reports by charts and graphs with KPI's. • It depends on a small scale of past data, and there is no intelligence involved; business management has to decide based on the information. • The quality of the solutions is volumetric and presents the accurate solution using data visualizations.  Features of Business Intelligence:	and 2 M for features

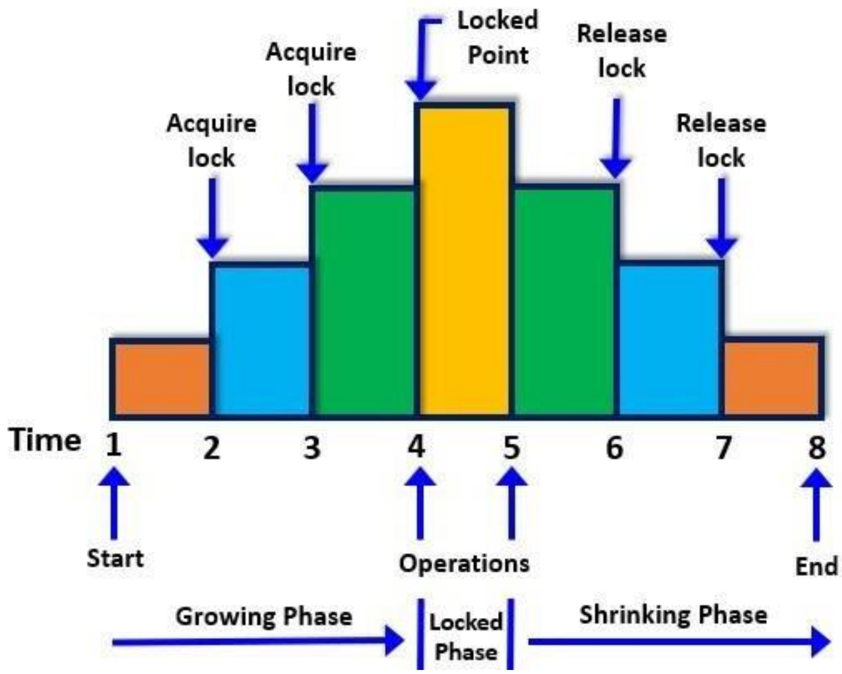
		1. **Data Warehouse**-Data warehouse includes the process of collecting all the data needed. In addition, data warehouse provides a data storage environment where data onto multiple data sources will be ETLed(Extracted, Transformed, Dunked) , cleaned up, and stored on a specific topic, indicating powerful data integration and maintenance capabilities of BI. 2. **Data Analysis**-You need the ability of data analysis to aid in enterprise modeling. OLAP is a data analysis tool based on data warehouse environment. Moreover, it solves the shortcoming of low efficiency of multi-dimensional analysis based on OLTP analysis. Business intelligence (BI) leverages data analysis to form actionable insights that inform an organization's strategic and tactical business decisions. 3. **Data Mining**-In practical applications, data mining is also used to mine the past and predict the future. Data mining is a deeper level of active design and analysis of business data. It is a process of using knowledge discovery tools to mine previously unknown and potentially useful knowledge. It is an active method of automatic discovery. 4. **Data Visualization**-Data visualization can reflect business operations intuitively. Enterprises could conduct purposeful analysis on those abnormal data and explore the possible causes. Then further make business decisions and strategies based on previous moves.	
	d)	**Describe oracle cloud & its features.**	**4 M**
	Ans	Oracle Cloud: • Oracle Cloud is one of the few cloud providers that can offer a complete set of cloud services to meet all your enterprise computing needs. • Use Oracle Infrastructure as a Service (IaaS) offerings to quickly set up the virtual machines, storage, and networking capabilities you need to run just about any kind of workload. Your infrastructure is managed, hosted, and supported by Oracle. • Use Oracle Platform as a Service offerings to provision ready-to-use environments for your enterprise IT and development teams, so they can build and deploy applications, based on proven Oracle databases and application servers. • Use Oracle Software as a Service (SaaS) offerings to run your business from the Cloud. Oracle offers cloud-based solutions for Human Capital Management, Enterprise Resource Planning, Supply Chain Management, and many other applications, all managed, hosted, and supported by Oracle. **Features of oracle Cloud:** • Oracle is completely scalable, as well as its style features a relatively distinct, scale-out style with smart storage space, flash technology.	2 M for Oracle cloud and 2 M for features of oracle cloud

		• Oracle is the selection of the majority of successful companies and enterprises as a result of its reliable data source monitoring alternatives and safer alternatives. • Capacities to enhance efficiency. • On-Demand Throughput and Storage Provisioning.	
5.		**Attempt any <u>TWO</u> of the following:**	**12 M**
	a)	**Explain flower expressions and Nested queries in XQuery.**	**6 M**
	Ans	The programming language XQuery defines **FLWOR** (pronounced 'flower') as an expression that supports iteration and binding of variables to intermediate results. **FLWOR** is an acronym: FOR, LET, WHERE, ORDER BY, RETURN. • **For** - selects a sequence of nodes • **Let** - binds a sequence to a variable • **Where** - filters the nodes • **Order by** - sorts the nodes • **Return** - what to return (gets evaluated once for every node) **E:g:** **for $x in doc("books.xml")/bookstore/book** **where $x/price>30** **order by $x/title** **return $x/title** The **for** clause selects all book elements under the bookstore element into a variable called $x. The **where** clause selects only book elements with a price element with a value greater than 30. The **order by** clause defines the sort-order. Will be sort by the title element. The **return** clause specifies what should be returned. Here it returns the title elements. **Nested XQuery** Nested XQuery can be considered the multiple nested operation from the XQuery. In nested queries, a query is written inside a query. The result of inner query is used in execution of outer query For each book in the bibliography, list the title and authors, grouped inside a result element. **E:g:** **<results>** **{** **for $b in doc("bib.xml")/bib/book**	3 M FLWOR Expressions explanation And 3 M explanation of Nested XQuery

		return **<result>** **{ $b/title }** **{** **for $a in $b/author** **return $a** **}** **</result>** **}** **</results>**	
	b)	**Write query to execute find () function on collection: Inventory.** **(i) to display all document where status equals "A" and "G".** **(ii) to display all document in collection.** **(iii) to display all documents where quality is less than 30 and greater than 10.**	**6 M**

	Ans	Query :-
		db.inventory.insertMany(
		[
		{
		First_Name: "Radhika",
		Last_Name: "Sharma",
		qty: 30,
		e_mail: "radhika_sharma.123@gmail.com",
		status: "B"
		},
		{
		First_Name: "Rachel",
		Last_Name: "Miller",
		qty: "15",
		e_mail: "Rachel_Christopher.123@gmail.com",
		status: "G"
		},
		{
		First_Name: "Fathima",
		Last_Name: "Sheik",
		qty: "25",
		e_mail: "Fathima_Sheik.123@gmail.com",
		status: "A"
		}
		]
		)
		i) db. inventory.find({$and : [{"status": "A"},{"status": "G"}]})
		ii) db.inventory.find()
			2 M for each correct query

		iii) db.inventory.find({$and: [{ “qty”: {$lt: 30}},{ “qty”: {$gt: 10}}}])	
	c)	**Explain inheritance for PL/SQL objects with examples**	**6 M**
	Ans	The object which inherits the behaviour of other objects is called as a Subtype and the object which is being inherited is called as the Supertype. When an instance is created for the subtype, the instance inherits the behaviour of both the supertype and the subtype. A subtype cannot inherit the properties of two supertypes when there is no relationship between them, similar to how a human child cannot have more than one father or one mother. But the subtype created on a supertype can inherit the properties of that supertype and all its parent supertypes if present. A method in a supertype is by default overridden when the subtype provides a method with the same name and type, similar to its supertype. **Example: -** CREATE TYPE food_ot AS OBJECT (name VARCHAR2(100), food_group VARCHAR2 (50), grown_in VARCHAR2 (100))NOT FINAL ; A very simple type: it contains just three attributes and no methods (procedures or functions). It also contains the NOT FINAL clause. This means, rather obviously, that I am "not done." What that means in the context of object types is that I might want to create a subtype of this type. CREATE TYPE dessert_t UNDER food_ot (contains_chocolate CHAR (1), year_created NUMBER(4)) NOT FINAL ; Notice the UNDER food_ot syntax in place of the IS OBJECT, creating a new type "under" food, it clearly is an object type.	3 M Explanation And 3 M Example
6.		**Attempt any <u>TWO</u> of the following:**	**12 M**

	a)	**Define lock. Explain two phase locking protocol with neat example.**	**6 M**
	Ans	**Locks:** Lock is a data variable which is associated with a data item. Locks help synchronize access to the database items by concurrent transactions. All lock requests are made to the concurrency-control manager. Transactions proceed **TWO PHASE LOCKING PROTOCOL: -**  **Two phase Locking protocol:** which is also known as 2PL. Two phase locking protocol requires that each transaction issues lock and unlock requests in two phases. **Growing phase:** A transaction may obtain locks but may not release any lock. **Shrinking phase:** A transaction may release locks, but may not obtain any new locks. If the conversion is allowed, then upgrading of locks from S(A) to X(A) happens in growing phase and the downgrade of locks from X(A) to S(A) happens in shrinking phase. It is true that 2PL protocol offers serializability. However, it does not ensure that dead locks not happen. Example: -	1 M definition 3 M explanation 2 M example

	T1	T1
1	Lock-S(A)	
2		Lock-S (A)
3	Lock- X(B)	
4	---	---
5	Unlock(A)	
6		Lock- X(C)
7	Unlock(B)	
8		Unlock(A)
9		Unlock(C)

This is just a skeleton transaction that shows how unlocking and locking work with 2-PL. Note for:

Transaction T1:

• The growing Phase is from steps 1-3.
• The shrinking Phase is from steps 5-7.
• Lock Point at 3

Transaction T2:

• The growing Phase is from steps 2-6.
• The shrinking Phase is from steps 8-9.
• Lock Point at 6

b) **Explain object & object identity. Write SQL query for the given table** **6 M**

Class: Employee
Name Salary Post Gender
Store Print Update

Ans

SQL objects :-
SQL objects are schemas, journals, catalogs, tables, aliases, views, indexes, constraints, triggers, sequences, stored procedures, user-defined functions, user-defined types, global variables, and SQL packages.
SQL creates and maintains these objects as system objects.

Object identity :-
Object identity is a fundamental object orientation concept. With object identity, objects can contain or refer to other objects.

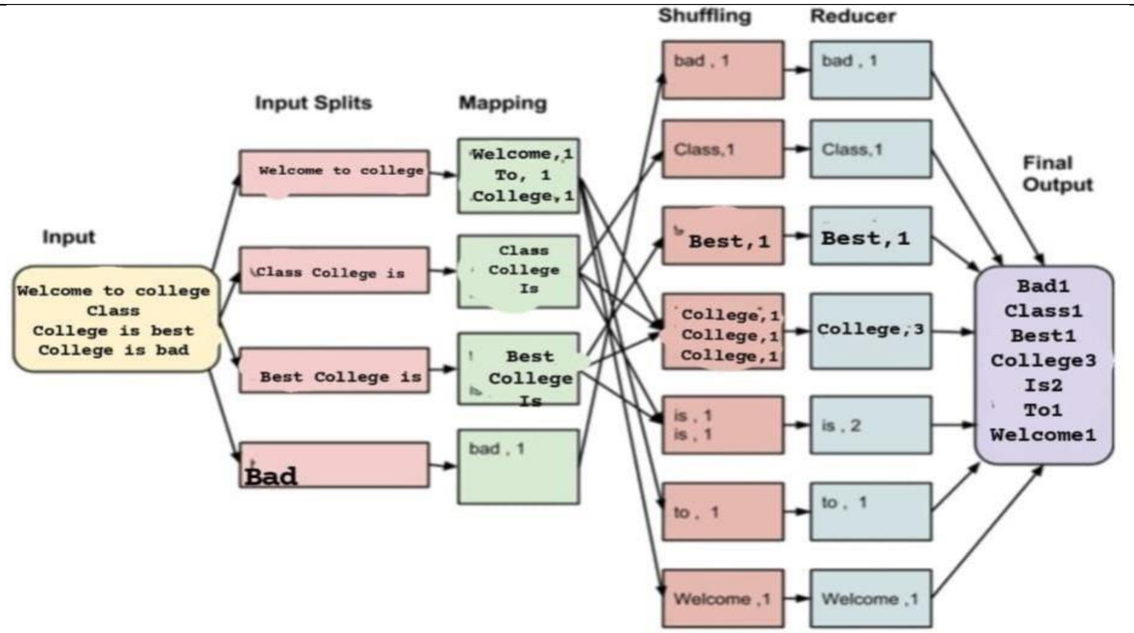
3 M explanation

3 M for correct query

		Identity is a property of an object that distinguishes the object from all other objects in the application.
		
		Query for given table :-
		
		CREATE TYPE person AS object(
		
		Name varchar(10),
		
		Post varchar(30),
		
		Salary number(10),
		
		Gender varchar(7))
		
		NOT FINAL
		
		METHOD print
		
		RETURNS text
		
		METHOD update
		
		RETURNS text;
		
		//method creation
		
		CREATE instance METHOD print
		RETURNS text
		FOR Person
		
		BEGIN
		RETURN self.name
		self.post
		self.salary
		self.gender;
		END
		
		CREATE instance METHOD update
		RETURNS text
		FOR Person
		BEGIN
		RETURN self.post=”Manager”;
		END
		
		// for output
		
		SELECT Name, print FROM Person;	

	c)	**Consider following input data for your Map Reduce Program:** **Welcome to college class** **College is best**	**6 M**

<table>
<tr><td></td><td></td><td>College is bad
Draw Map Reduce Architecture and explain its phases.</td><td></td></tr>
<tr><td></td><td>Ans</td><td>



Phases of Map Reduce are as follows:-

Input Files: -The data that is to be processed by the MapReduce task is stored in input files. These input files are stored in the Hadoop Distributed File System. The file format is arbitrary, while the line-based log files and the binary format can also be used.

Input Splits: An input in the MapReduce model is divided into small fixed-size parts called input splits. This part of the input is consumed by a single map. The input data is generally a file or directory stored in the HDFS.

Mapping: This is the first phase in the map-reduce program execution where the data in each split is passed line by line, to a mapper function to process it and produce the output values.

Shuffling: It is a part of the output phase of Mapping where the relevant records are consolidated from the output. It consists of merging and sorting. So, all the key-value pairs which have the same keys are combined. In sorting, the inputs from the merging step are taken and sorted. It returns key-value pairs, sorting the output.

Reduce: All the values from the shuffling phase are combined and a single output value is returned. Thus, summarizing the entire dataset.

</td><td>3 M for map reduce architecture

3 M for explanation</td></tr>
</table>